

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

PMIC for enhanced dual-source energy harvesting efficiency

Recently, e-peas has launched a power management integrated circuit (PMIC). The AEM13920 enhances the potential of energy harnessing by adeptly managing inputs from any two sources concurrently, which could be a mix of photovoltaic (PV) cells, thermo-electric



generators (TEGs), RF energy harvesters, or intermittent energy sources. The PMIC boasts

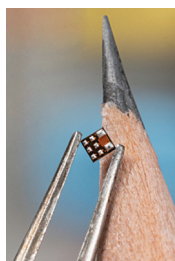
remarkably high conversion efficiencies, exceeding 90%, from both source-to-storage and storage-to-load, facilitating the maximum energy extraction from any kind of harvester. Employing the PMIC streamlines the process of power system development, thereby reducing both the number of components and the overall cost of materials. Moreover, a distinct buck regulator on the chip alters the output from the storage segment to a set voltage, ranging between 0.6V and 2.5V, as mandated by the application.

e-peas
<https://e-peas.com>

Smallest current sensor for simplified design and increased accuracy

Texas Instruments (TI) has launched a current sensor to help engineers simplify their designs while improving accuracy. The sensor features the lowest-drift isolated Hall-effect current for high-voltage systems, coupled with a collection of current shunt monitors

that eliminate the need for an external shunt resistor for non-isolated voltage rails. The TMCS1123 brings a blend of design ease and precision for high-voltage systems, boasting unmatched reinforced isolation and unparalleled accuracy throughout its lifespan and



across various temperatures in the sector. The TMCS1123 stands out with a low propagation delay of 600ns and a bandwidth of 250kHz, facilitating quicker control loops and maintaining minimal noise, thereby augmenting system efficiency.

Texas Instruments
<https://www.ti.com>

A complete voice solution revolutionising power and flexibility

XMOS has launched xcore-voice, a complete voice solution built on xcore platform hardware, software, and tools. The solution provides voice pipeline designs, utilising XMOS's



tried-and-tested audio front-end, which features far-field processing and compatibility with third-

party independent software vendor (ISV) voice algorithms. The solution facilitates product designers in swiftly and economically introducing voice interfaces without compromising audio quality. The platform stands out with its low-power architecture, drastically reducing power consumption to less than 55mW, aligning with

EU energy-saving norms, and offering an 'always-on' feature in modern electronics. It provides application-specific software components, including modifiable algorithms and core components like FreeRTOS for optimal utilisation. Furthermore, it integrates general-purpose digital signal processing (DSP), artificial intelligence (AI), and input/output (I/O) processing, customisable via software, streamlining design paths and offering a BOM-optimised solution for fast integration.

XMOS
<https://www.xmos.ai>

Customisable powerhouse for faster computing

Semidynamics has launched the Atrevido 423 featuring a broader 4-way pipeline, doubling the decoding and retirement capacity compared to its recently unveiled 2-way, 223 core. The core presents an out-of-order framework,



extending a variety of RISC-V extensions that can be incorporated. A standout feature is the option to equip it with the proprietary vector

unit, adhering fully to the latest RISC-V vector specifications. For enhanced security, customers have the option to safeguard the data cache with ECC and the instruction cache with parity. The core presents a unique feature where the bit number of each vector register can be customised to suit the client's needs, promoting efficiency in handling large memory latencies and power reduction. This innovation enables the vector unit to process an unparalleled amount of data, continually fed by the Gazillion.

Semidynamics
<https://semidynamics.com>